Database Design

Agenda

OBJECT™ TECHNOLOGIES

1. Data Models
2. ACID Properties
3. Concepts of database instance and schema
4. MySQL storage engines
5. Summary

1. Data Models

OBJECT™ TECHNOLOGIES

Following are the data models

- **Hierarchical Model**
  - -Tree like structure
  - -Index of a book, recipes etc
  - -One-to-many relationship
    - e.g 1 dept-many professors

- **Network Model**
  - -Extension of the Hierarchical model
  - -Data is organized more like a graph
  - -More than one parent node
  - -Accessing the data is easier and fast.
  - -Model was used to map many-to-many data relationships

- **Entity-relationship Model**
  - -High-level conceptual data model
  - -Systematically analyze data requirements to produce a well-designed database
  - -Entities/Attributes/Relationship

- **Relational Model**
  - -Introduced by E.F Codd in 1970
  - -tables are also known as relations
  - -Oracle, MySQL, PostgreSql, MariaDB

2. ACID Properties

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- **Atomicity**
  - Atomicity implies that either whole entire transaction takes place at once or nothing.
- **Consistency**
  - Data accessed from multiple sources should always be same.
  - Database is consistent before and after the transaction is done.
- **Isolation**
  - Data isolation implies that the changes made by any transaction will not be reflected unless entire transaction is completed successfully.
  - All other dependent transactions are kept isolated till the current transaction completes.
  - This property ensures that multiple transactions can occur concurrently without leading to the inconsistency of database state.
- **Durability**
  - Last successful transaction should be reflected even if the system fails

The ACID properties, provide a mechanism to ensure correctness and consistency of a database in a way such that each transaction is a group of operations that acts as a single unit, produces consistent results, acts in isolation from other operations and updates that it makes are stored properly.

3. Concepts of database instance and schema

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- **Database Schema**
  - Abstract design that represents the storage of your data in a database.
  - It describes both the organization of data and the relationships between tables in a given database
  - Developers plan a database schema in advance.
  - Blueprint or architecture of how our data will look.
  - It dose not hold data.
- **Database Instance**
  - Instances are the collection of information stored at a particular moment.
  - The situation where a data or information is stored in the database at a particular moment of time is called an instance.
  - Instances can be changed by certain operations as like addition, deletion of data.
  - Search query will not make any kind of changes in the instances.
  - An instance is also called a current state or database state.
- **Subschema**
  - A subschema provides a view of the database as seen by an application program.

4. MySQL storage engines

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- **InnoDB**
  - Default storage engine of MySQL since 5.5 and above.
  - Provides ACID compliant tables, supports FK, commit, rollback, crash recovery, provides multiuser environment, stores data using clustered index.
- **MyISAM**
  - Manage non transactional tables, provides high speed storage and retrieval. Supports full text searching.
- **Memory**
  - The MEMORY tables store data entirely in memory and use hash indexes so that they are faster than MyISAM tables.
  - Initially called as HEAP.
- **MERGE**
  - Groups more than one similar MyISAM tables to be treated as a single table. Can handle non transactional tables.
  - Included by default.
- **Eamples**
  - Can create tables.
  - Data cannot be fetched.
  - Purpose is to teach developers about how to write a storage engine.
- **ARCHIVE**
  - Stores large amount of data.
  - Dose not support index.
- **CSV**
  - The CSV storage engine stores data in comma-separated values (CSV) file format. A CSV table brings a convenient way to migrate data into non-SQL.
- **BLACKHOLE**
  - Accepts data to store but always returns empty.
- **FEDERATED**
  - Stores data in remote database.

Summary

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